TASK -2 LINUX COMMANDS

CHAPTER – 1 - SYSTEM INFORMATION

<1>uname:

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
gopika_arumugm@cloudshell:~$ uname
Linux
```

<2>uptime:

```
gopika_arumugm@cloudshell:~$ uptime
 07:29:51 up  2:35,  0 user,  load average: 0.10, 0.04, 0.01
gopika_arumugm@cloudshell:~$
```

<3> hostname –

```
gopika_arumugm@cloudshell:~$ hostname
cs-553674751103-default
```

<4> dmesg –

```
gopika_arumugm@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
gopika_arumugm@cloudshell:~$
```

<5> free –

```
gopika_arumugm@cloudshell:~$ free
               total        used        free      shared  buff/cache   available
Mem:        16382844     1486600    13239924        1136     1940976    14896244
Swap:              0           0           0
gopika_arumugm@cloudshell:~$
```

<6> top –

```
top - 07:32:14 up  2:38,  0 user,  load average: 0.01, 0.02, 0.00
Tasks:  42 total,   1 running,  41 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  2.4 sy,  0.0 ni, 97.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15998.9 total,  12928.1 free,   1453.2 used,   1895.5 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.  14545.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0    4324   3256   2976 S   0.0   0.0   0:00.09 bash
      9 syslog    20   0  222140   3912   2552 S   0.0   0.0   0:00.06 rsyslogd
     25 root      20   0   38768  27792  10688 S   0.0   0.2   0:01.69 python
     26 root      20   0    6196   2680   2440 S   0.0   0.0   0:00.00 logger
     67 root      10 -10   12024   4252   3140 S   0.0   0.0   0:00.07 sshd
    388 root      20   0 2096096  85432  61664 S   0.0   0.5   0:01.36 dockerd
    442 root      20   0 1234524  13332   8212 S   0.0   0.1   0:01.80 editor-proxy
    443 root      20   0   16328   6424   5500 S   0.0   0.0   0:00.01 sudo
    456 root      20   0 1226100   2764   1988 S   0.0   0.0   0:00.01 tmux-agent
    495 root      20   0 1867468  45956  32032 S   0.0   0.3   0:09.28 containerd
    515 root      10 -10   13996   9708   8328 S   0.0   0.1   0:00.02 sshd
    586 gopika_+  10 -10   14384   6512   4788 S   0.0   0.0   0:00.02 sshd
    652 gopika_+  10 -10    7340   3804   3552 S   0.0   0.0   0:00.00 bash
    656 gopika_+  10 -10    7604   4428   3836 S   0.0   0.0   0:00.00 bash
    705 root      20   0    2696   1576   1476 S   0.0   0.0   0:00.00 sleep
    711 gopika_+  10 -10    7956   3604   2212 S   0.0   0.0   0:00.13 tmux
    712 gopika_+  10 -10    8800   5832   3988 S   0.0   0.0   0:00.05 bash
    715 root      20   0    9196   4044   3668 S   0.0   0.0   0:00.00 runuser
    721 gopika_+  20   0    4324   3236   2972 S   0.0   0.0   0:00.00 sh
    758 gopika_+  20   0   11.3g 152092  51048 S   0.0   0.9   0:11.06 node
   1192 gopika_+  20   0 1048516  61912  47496 S   0.0   0.4   0:00.99 node
   1525 gopika_+  20   0   42.4g 686024  54524 S   0.0   4.2   0:20.98 node
   1564 gopika_+  20   0 1016648  62816  46104 S   0.0   0.4   0:00.64 node
   1723 root      10 -10   13996   9708   8336 S   0.0   0.1   0:00.01 sshd
   1725 gopika_+  10 -10   14256   6376   4728 S   0.0   0.0   0:00.03 sshd
   1726 gopika_+  10 -10    7340   3612   3364 S   0.0   0.0   0:00.00 bash
   1727 gopika_+  10 -10    7340   3828   3568 S   0.0   0.0   0:00.00 start-shell.sh
   1728 gopika_+  10 -10    6880   3384   3080 S   0.0   0.0   0:00.00 tmux
   1729 gopika_+  10 -10    8800   5908   4056 S   0.0   0.0   0:00.07 bash
   2207 root      10 -10   13996   9796   8420 S   0.0   0.1   0:00.02 sshd
   2209 gopika_+  10 -10   14256   6376   4720 S   0.0   0.0   0:00.02 sshd
```

<7> htop –

```
    0[||                                                                                     1.0%] Tasks: 42, 69 thr, 0 kthr; 1 running
    1[|                                                                                      0.5%] Load average: 0.01 0.02 0.00
    2[||                                                                                     1.6%] Uptime: 02:40:39
    3[||                                                                                     1.6%]
  Mem[||||||||||||||||||||                                                            1.16G/15.6G]
  Swp[                                                                                      0K/0K]

  Main  I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
    452 root        20   0 1205M 13332  8212 S   0.0  0.1  0:00.31 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
    453 root        20   0 1205M 13332  8212 S   0.0  0.1  0:00.05 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
    454 root        20   0 1205M 13332  8212 S   0.0  0.1  0:00.35 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
    456 root        20   0 1197M  2764  1988 S   0.0  0.0  0:00.00 /google/devshell/tmux-agentn/docker/containerd/containerd.toml
    464 root        20   0 1197M  2764  1988 S   0.0  0.0  0:00.00 /google/devshell/tmux-agentn/docker/containerd/containerd.toml
    467 root        20   0 1197M  2764  1988 S   0.0  0.0  0:00.00 /google/devshell/tmux-agent
    468 root        20   0 1197M  2764  1988 S   0.0  0.0  0:00.00 /google/devshell/tmux-agentn/docker/containerd/containerd.toml
    469 root        20   0 1197M  2764  1988 S   0.0  0.0  0:00.00 /google/devshell/tmux-agentn/docker/containerd/containerd.tomlpervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/
    487 root        20   0 1205M 13332  8212 S   0.0  0.1  0:00.21 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
    495 root        20   0 1823M 45956 32032 S   0.0  0.3  0:00.06 containerd --config /var/run/docker/containerd/containerd.tomlmirror=https://asia-mirror.gcr.io10-60 startups
    496 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    417 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.22 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    418 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.19 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    419 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    420 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    421 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    440 root        20   0 2046M 85432 61664 S   0.0  0.5  0:00.17 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
    778 gopika_aru  20   0 11.3G  148M 51048 S   0.0  0.9  0:00.16 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.js --ho
    779 gopika_aru  20   0 11.3G  148M 51048 S   0.0  0.9  0:00.16 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.js --ho
    780 gopika_aru  20   0 11.3G  148M 51048 S   0.0  0.9  0:00.16 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.js --ho
    781 gopika_aru  20   0 11.3G  148M 51048 S   0.0  0.9  0:00.16 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.js --ho
    950 root        20   0 1205M 13332  8212 S   0.0  0.1  0:00.32 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
   1192 gopika_aru  20   0 1023M 61924 47496 S   0.0  0.4  0:00.66 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-fork --ty
   1193 gopika_aru  20   0 1023M 61924 47496 S   0.0  0.4  0:00.00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-fork --ty
   1194 gopika_aru  20   0 1023M 61924 47496 S   0.0  0.4  0:00.00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-fork --ty
   1195 gopika_aru  20   0 1023M 61924 47496 S   0.0  0.4  0:00.01 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-fork --ty
   1196 gopika_aru  20   0 1023M 61924 47496 S   0.5  0.4  0:00.07 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-fork --ty
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

<8>lsb_release –

```
gopika_arumugm@cloudshell:~$ lsb_release
No LSB modules are available.
gopika_arumugm@cloudshell:~$
```

<9> arch –

```
gopika_arumugm@cloudshell:~$ arch
x86_64
gopika_arumugm@cloudshell:~$
```

CHAPTER – 2 FILE AND DIRECTORY MANAGEMENT

<1> ls -

```
gopika_arumugm@cloudshell:~$
gopika_arumugm@cloudshell:~$ ls
admin  README-cloudshell.txt  user
gopika_arumugm@cloudshell:~$
```

<2> cd -

```
gopika_arumugm@cloudshell:~$ cd user
gopika_arumugm@cloudshell:~/user$
```

<3> pwd -

```
gopika_arumugm@cloudshell:~$ cd user
gopika_arumugm@cloudshell:~/user$ pwd
/home/gopika_arumugm/user
```

<4> mkdir -

```
/home/gopika_arumugm/user
gopika_arumugm@cloudshell:~/user$ mkdir mlops
gopika_arumugm@cloudshell:~/user$
```

<5> rmdir -

```
gopika_arumugm@cloudshell:~/user$ ls
mlops
gopika_arumugm@cloudshell:~/user$ rmdir mlops
gopika_arumugm@cloudshell:~/user$ ls
gopika_arumugm@cloudshell:~/user$
```

<6> touch -

```
gopika_arumugm@cloudshell:~/user$ touch f.txt
gopika_arumugm@cloudshell:~/user$ ls
f.txt
```

<7> rm -

```
gopika_arumugm@cloudshell:~/user$ rm f.txt
gopika_arumugm@cloudshell:~/user$ ls
gopika_arumugm@cloudshell:~/user$
```

<8> cp -

```
gopika_arumugm@cloudshell:~/user$ echo "Hello World"> file.txt
gopika_arumugm@cloudshell:~/user$ ls
file.txt
gopika_arumugm@cloudshell:~/user$ cp file.txt nw.txt
gopika_arumugm@cloudshell:~/user$ ls
file.txt  nw.txt
gopika_arumugm@cloudshell:~/user$ cat nw.txt
Hello World
```

<9> mv -

```
gopika_arumugm@cloudshell:~/user$ mv file.txt nwfile.txt
gopika_arumugm@cloudshell:~/user$ ls
nwfile.txt  nw.txt
gopika_arumugm@cloudshell:~/user$
```

<10> ln -

```
gopika_arumugm@cloudshell:~/user$ ln nw.txt hello.txt
gopika_arumugm@cloudshell:~/user$ ls
hello.txt  nwfile.txt  nw.txt
```

CHAPTER – 3 FILE VIEWING AND EDITING

<1> cat -

```
gopika_arumugm@cloudshell:~/user$ cat hello.txt
Hello World
gopika_arumugm@cloudshell:~/user$ more hello.txt
Hello World
```

<2> more -

```
ent systems. One of the most important features of Linux is its command-line interface, which allows users to perform t
asks efficiently using commands instead of graphical tools. Among these commands, file handling commands play a very im
portant role because everything in Linux is treated as a file, including directories, devices, and system resources.

File operations in Linux include creating, reading, writing, copying, moving, and deleting files. To create a file, com
mands like touch or echo are used. The touch command creates an empty file, while echo can be used to write content int
o a file using redirection operators like > and >>. The > operator is used to overwrite content in a file, whereas >> i
s used to append data to an existing file without deleting previous content.

--More--(37%)
```

<3> less –

```
Hello World
Linux is a powerful open-source operating system that is widely used in servers, programming environments, and development systems. One of the most important features of Linux is its command-li
ne interface, which allows users to perform tasks efficiently using commands instead of graphical tools. Among these commands, file handling commands play a very important role because everythi
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xisting file without deleting previous content.

To view the contents of a file, commands such as cat, more, and less are used. The cat command displays the entire content at once, which is useful for small files. However, for larger files, m
ore and less are more helpful because they allow users to scroll through the content page by page. The more command shows content one screen at a time, while less provides additional navigation
 options such as moving backward and searching within the file.

Linux also provides file management commands like cp, mv, and rm. The cp command is used to copy files from one location to another, while mv is used to move or rename files. The rm command is
used to delete files permanently. These commands are very powerful and should be used carefully because Linux does not have a recycle bin by default.

Another important concept in Linux is file linking, which is done using the ln command. A hard link creates another name for an existing file, meaning both names point to the same data on the d
isk. A symbolic link, created using ln -s, works like a shortcut to the original file. If the original file is deleted, the symbolic link will stop working, but the hard link will still remain
valid.

Overall, Linux file commands are essential for system navigation, programming, and development tasks. Understanding these commands helps users work efficiently in a terminal environment and imp
roves control over the operating system. Mastering file handling in Linux is one of the first steps toward becoming proficient in system administration or software development
hello.txt (END)
```

<4> tail

```
gopika_arumugm@cloudshell:~/user$ tail hello.txt

File operations in Linux include creating, reading, writing, copying, moving, and deleting files. To create a file, com
mands like touch or echo are used. The touch command creates an empty file, while echo can be used to write content int
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ering file handling in Linux is one of the first steps toward becoming proficient in system administration or software
development
```

<5> head

```
gopika_arumugm@cloudshell:~/user$ head hello.txt
Hello World
Linux is a powerful open-source operating system that is widely used in servers, programming environments, and developm
ent systems. One of the most important features of Linux is its command-line interface, which allows users to perform t
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rovides additional navigation options such as moving backward and searching within the file.
```

<6>nano

```
gopika_arumugm@cloudshell:~/user$ head hello.txt
Hello World
Linux is a powerful open-source operating system that is widely used in servers, pr
  GNU nano 7.2                           hello.txt *
>> is used to append data to an existing file without deleting previous
content.

To view the contents of a file, commands such as cat, more, and less areving, a>
used. The cat command displays the entire content at once, which is
useful for small files. However, for larger files, more and less are  re used. >
more helpful because they allow users to scroll through the content page
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provides additional navigation options such as moving backward and
searching within the file.
echo are used. The touch command creates an empty file, while echo can
be used to write content into a file using redirection operators like >amming, >
and >>. The > operator is used to overwrite content in a file, whereas
Another important concept in Linux is file linking, which is done using the ln >
content.
Overall, Linux file commands are essential for system navigation, programming, >
To view the contents of a file, commands such as cat, more, and less are used.

Linux also provides file management commands like cp, mv, and rm. The cp comman>

                              [ Justified paragraph ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

<7>vi/vm –

```
gopika_arumugm@cloudshell:~/user$ vi hello.txt
gopika_arumugm@cloudshell:~/user$ cat hello.txt
Updated using vi:

To view the contents of a file, commands such as cat, more, and less are used. The cat command displays the entire con
tent at once, which is useful for small files. However, for larger files, more and less are more helpful because they
allow users to scroll through the content page by page. The more command shows content one screen at a time, while les
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Overall, Linux file commands are essential for system navigation, programming, and development tasks. Understanding th
ese commands helps users work efficiently in a terminal environment and improves control over the operating system. Ma
stering file handling in Linux is one of the first steps toward becoming proficient in system administration or softwa
re development
gopika_arumugm@cloudshell:~/user$
```

<8> gedit – NA

<9> awk –

```
gopika_arumugm@cloudshell:~/user$ awk '{print$1}' struct.txt
Name
Ravi
Asha
John
Meena
Suresh
Anita
Vikram
Priya
gopika_arumugm@cloudshell:~/user$ awk "NR<=3" struct.txt
Name Age Marks City
Ravi 20 85 Lucknow
Asha 19 92 Kanpur
gopika_arumugm@cloudshell:~/user$
```

<10> sed –

```
gopika_arumugm@cloudshell:~/user$ sed 's/Ravi/RaviKumar/' struct.txt
Name Age Marks City
RaviKumar 20 85 Lucknow
Asha 19 92 Kanpur
John 21 78 Delhi
Meena 20 88 Lucknow
Suresh 22 75 Jaipur
Anita 19 95 Kanpur
```

CHAPTER – 4 FILE PERMISSIONS AND OWNERSHIPS

<1> chmod –

```
gopika_arumugm@cloudshell:~$ echo "Hello World" file.txt
Hello World file.txt
gopika_arumugm@cloudshell:~$ chmod -w file.txt
```

<2> chown-

```
gopika_arumugm@cloudshell:~$ ^C
gopika_arumugm@cloudshell:~$ chown gopika_arumugm file.txt
```

<3> chgrp

```
gopika_arumugm@cloudshell:~$ groups
gopika_arumugm adm sudo docker
gopika_arumugm@cloudshell:~$ chgrp sudo file.txt
gopika_arumugm@cloudshell:~$
```

<4>umask

```
gopika_arumugm@cloudshell:~$ umask
0002
gopika_arumugm@cloudshell:~$
```

<5> stat

```
gopika_arumugm@cloudshell:~$ stat file.txt
  File: file.txt
  Size: 12              Blocks: 8          IO Block: 4096   regular file
Device: 8,17    Inode: 131194      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/gopika_arumugm)   Gid: (   27/    sudo)
Access: 2026-05-14 23:43:07.903441928 +0000
Modify: 2026-05-15 00:20:48.447027795 +0000
Change: 2026-05-15 00:22:30.784706951 +0000
 Birth: 2026-05-14 15:13:06.584384142 +0000
```

<6>getfacl

```
gopika_arumugm@cloudshell:~$ getfacl file.txt
# file: file.txt
# owner: gopika_arumugm
# group: sudo
user::rw-
group::rw-
other::r--
```

<7>setfacl

```
gopika_arumugm@cloudshell:~$ getfacl file.txt
# file: file.txt
# owner: gopika_arumugm
# group: sudo
user::rw-
group::rw-
mask::rw-
other::r--
```

<8>lsattr

```
gopika_arumugm@cloudshell:~$ lsattr file.txt
--------------e------- file.txt
```

<9> chattr

```
gopika_arumugm@cloudshell:~$ sudo chattr +i file.txt
gopika_arumugm@cloudshell:~$ lsattr file.txt
----i---------e------- file.txt
```

<10>usermod

```
gopika_arumugm@cloudshell:~$ ^C
gopika_arumugm@cloudshell:~$ sudo usermod -l gops gopika_arumugm
```

CHAPTER – 5 NETWORKING COMMANDS

<1> ipconfig

```
gopika_arumugm@cloudshell:~$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1460
        inet 10.88.0.3  netmask 255.255.0.0  broadcast 10.88.255.255
        ether be:a3:2e:be:75:db  txqueuelen 1000  (Ethernet)
        RX packets 32045  bytes 30330150 (30.3 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 28027  bytes 24157997 (24.1 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

<2> ip

```
gopika_arumugm@cloudshell:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen
 1000
    link/ether be:a3:2e:be:75:db brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.3/16 brd 10.88.255.255 scope global eth0
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 3a:a4:6e:e4:77:39 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

<3> ping

```
gopika_arumugm@cloudshell:~$ ping google.com
PING google.com (172.217.194.101) 56(84) bytes of data.
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=1 ttl=114 time=0.654 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=2 ttl=114 time=0.458 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=3 ttl=114 time=0.397 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=4 ttl=114 time=0.420 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=5 ttl=114 time=0.430 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=6 ttl=114 time=0.385 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=7 ttl=114 time=0.382 ms
64 bytes from si-in-f101.1e100.net (172.217.194.101): icmp_seq=8 ttl=114 time=0.431 ms
```

<4>netstat

```
gopika_arumugm@cloudshell:~$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:45815         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:971             0.0.0.0:*               LISTEN
tcp        0      0 localhost:34383         0.0.0.0:*               LISTEN
tcp        0      0 localhost:8998          0.0.0.0:*               LISTEN
tcp        0      0 localhost:44391         0.0.0.0:*               LISTEN
tcp        0      0 localhost:ssh           localhost:55692         ESTABLISHED
tcp        0      0 localhost:41332         localhost:ssh           TIME_WAIT
tcp        0      0 localhost:55928         localhost:ssh           ESTABLISHED
tcp        0      0 localhost:971           localhost:43738         ESTABLISHED
tcp        0      0 cs-553674751103-d:45034 sk-in-f101.1e100.:https ESTABLISHED
tcp        0      0 localhost:45294         localhost:970           ESTABLISHED
tcp        0      0 localhost:58790         localhost:ssh           TIME_WAIT
```

<5> ss

```
gopika_arumugm@cloudshell:~$ ss -l
Netid State   Recv-Q Send-Q                                                        Local Address:Port
                         Peer Address:Port                                    Process
nl     UNCONN 0       0                                                                 rtnl:kernel
                                      *
nl     UNCONN 960     0                                                                 rtnl:dockerd/352
                                      *
nl     UNCONN 960     0                                                              tcpdiag:kernel
                                      *
nl     UNCONN 4352    0                                                              tcpdiag:ss/2983
                                      *
nl     UNCONN 0       0                                                                 xfrm:kernel
                                      *
nl     UNCONN 0       0                                                                 xfrm:dockerd/352
                                      *
nl     UNCONN 0       0                                                                audit:kernel
                                      *
nl     UNCONN 0       0                                                                audit:sudo/438
                                      *
```

<6>traceroute

```
gopika_arumugm@cloudshell:~$ traceroute google.com
traceroute to google.com (74.125.130.102), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.059 ms  0.013 ms  0.010 ms
 2  * * *
 3  66.249.95.194 (66.249.95.194)  1.370 ms 72.14.232.246 (72.14.232.246)  1.378 ms *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
```

<7>wget

```
gopika_arumugm@cloudshell:~$ wget https://www.google.com/robots.txt
--2026-05-15 00:44:44--  https://www.google.com/robots.txt
Resolving www.google.com (www.google.com)... 142.251.151.119, 142.251.156.119, 142.251.155.119, .
..
Connecting to www.google.com (www.google.com)|142.251.151.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 6549 (6.4K) [text/plain]
Saving to: ‘robots.txt’

robots.txt               100%[===============================>]   6.40K  --.-KB/s    in 0.008s

2026-05-15 00:44:44 (771 KB/s) - ‘robots.txt’ saved [6549/6549]
```

<8>curl

```
gopika_arumugm@cloudshell:~$ curl https://www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en-SG"><head><meta c
ontent="text/html; charset=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/googl
eg/1x/googleg_standard_color_128dp.png" itemprop="image"><title>Google</title><script nonce="xG7a
gRhcBsdAWwhgcBZC3A">(function(){var _g={kEI:'gWwGaqncKtjE4-EP8ty5wAs',kEXPI:'0,824381,2232,477590
,3030760,344796,5520325,12270,11,193,296,36811273,25228681,217553,114375,61769,77237,30411,7714,3
3385,29820,22001,10371,35968,32561,11388,4791,2646,13253,10993,1472,2,14414,3,1515,3355,197,5813,
2,5999,16,2669,11622,2,876,2229,5089,19,3008,27,768,36623,5337,845,2046,2,13,8516,1745,4,17977,5,
964,2,147,389,4,2647,5,5568,298,6256,2997,3066,7,6984,6436,2,6335,1649,4,12676,2,4981,10,814,2,37
68,1638,544,4,4000,187,2105,5,461,3806,119,4,2364,2363,5,2609,4,3845,4,1232,818,371,5,1944,5,3772
,1144,4,2703,4,40,4,2110,4,438,2676,1906,5,1185,4,1201,5,75,4,1940,4,1649,4,1341,1,1357,2701,4,47
```

<9>nslookup

```
Non-authoritative answer:
Name:   google.com
Address: 64.233.170.113
Name:   google.com
Address: 64.233.170.139
Name:   google.com
Address: 64.233.170.101
Name:   google.com
Address: 64.233.170.100
Name:   google.com
Address: 64.233.170.138
Name:   google.com
Address: 64.233.170.102
Name:   google.com
Address: 2404:6800:4003:c02::66
Name:   google.com
Address: 2404:6800:4003:c02::64
Name:   google.com
Address: 2404:6800:4003:c02::8a
Name:   google.com
Address: 2404:6800:4003:c02::8b
```

<10>dig

```
gopika_arumugm@cloudshell:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26075
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             295     IN      A       64.233.170.113
google.com.             295     IN      A       64.233.170.139
google.com.             295     IN      A       64.233.170.101
```

CHAPTER – 6 PROCESS MANAGEMENT

<1> ps -

```
gopika_arumugm@cloudshell:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 May14 ?        00:00:00 /bin/bash /google/scripts/onrun.sh sleep infi
syslog         9       1  0 May14 ?        00:00:00 /usr/sbin/rsyslogd
root          25       1  0 May14 ?        00:00:01 /usr/bin/python /usr/bin/supervisord -n -c /g
root          26       1  0 May14 ?        00:00:00 logger -t supervisord
root          67       1  0 May14 ?        00:00:00 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysF
root         352       1  0 May14 ?        00:00:00 /usr/bin/dockerd -p /var/run/docker.pid --mtu
root         437      25  0 May14 ?        00:00:00 /google/devshell/editor/proxy/editor-proxy -s
root         438      25  0 May14 ?        00:00:00 sudo /google/devshell/tmux-agent
root         445     438  0 May14 ?        00:00:00 /google/devshell/tmux-agent
root         463     352  0 May14 ?        00:00:06 containerd --config /var/run/docker/container
root         521     437  0 May14 ?        00:00:00 runuser -l gopika_arumugm -c export GOPATH=/h
gopika +     522     521  0 May14 ?        00:00:00 sh ./bin/codeoss-cloudshell --host=0.0.0.0 --
```

<2> pgrep -

```
gopika_arumugm@cloudshell:~$ pgrep bash
1
704
1732
1735
1750
```

<3> pkill -

```
gopika_arumugm@cloudshell:~$ pkill firefox
gopika_arumugm@cloudshell:~$
```

<4> kill -

```
gopika_arumugm@cloudshell:~$ kill 1234
-bash: kill: (1234) - No such process
```

<5> killall -

```
gopika_arumugm@cloudshell:~$ killall firefox
firefox: no process found
```

<6> bg –

```
gopika_arumugm@cloudshell:~$ bg
[1]+ sleep 100 &
```

<7> fg -

```
gopika_arumugm@cloudshell:~$ fg
sleep 100
```

<8> jobs -

```
gopika_arumugm@cloudshell:~$ jobs
gopika_arumugm@cloudshell:~$
```

<9> nice -

```
gopika_arumugm@cloudshell:~$ nice -n 10 sleep 100

^C
gopika_arumugm@cloudshell:~$ ps -l
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000     704     702  0  70 -10 -  2231 do_wai pts/2    00:00:00 bash
0 R  1000    3208     704  0  70 -10 -  2730 -      pts/2    00:00:00 ps
```

<10> renice –

```
gopika_arumugm@cloudshell:~$ renice 5 -p 704
704 (process ID) old priority -10, new priority 5
```

CHAPTER – 7 DISK MANAGEMENT

<1> df –

```
gopika_arumugm@cloudshell:~$ df -h
Filesystem                          Size  Used Avail Use% Mounted on
overlay                              95G   73G   22G  78% /
tmpfs                                64M     0   64M   0% /dev
/dev/sda1                            95G   73G   22G  78% /root
/dev/disk/by-id/google-home-part1   4.8G  151M  4.4G   4% /home
/dev/root                           2.0G  1.3G  715M  64% /usr/lib/modules
shm                                  64M     0   64M   0% /dev/shm
gopika_arumugm@cloudshell:~$
```

<2> du -

```
gopika_arumugm@cloudshell:~$ du -sh .
151M    .
```

<3> fdisk -

```
Disk model: PersistentDisk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: gpt
Disk identifier: 88CD0047-6E3D-984A-9279-9D30D1B6BC8F

Device        Start       End   Sectors  Size Type
/dev/sda1   8704000 209715166 201011167 95.8G Linux filesystem
/dev/sda2     20480     53247     32768   16M ChromeOS kernel
/dev/sda3   4509696   8703999   4194304    2G ChromeOS root fs
/dev/sda4     53248     86015     32768   16M ChromeOS kernel
/dev/sda5    315392   4509695   4194304    2G ChromeOS root fs
/dev/sda6     16448     16448         1  512B ChromeOS kernel
/dev/sda7     16449     16449         1  512B ChromeOS root fs
/dev/sda8     86016    118783     32768   16M Linux filesystem
/dev/sda9     16450     16450         1  512B ChromeOS reserved
/dev/sda10    16451     16451         1  512B ChromeOS reserved
/dev/sda11       64     16447     16384    8M BIOS boot
/dev/sda12   249856    315391     65536   32M EFI System

Partition 7 does not start on physical sector boundary.
Partition 9 does not start on physical sector boundary.
Partition 10 does not start on physical sector boundary.
Partition table entries are not in disk order.


Disk /dev/dm-0: 1.94 GiB, 2087714816 bytes, 509696 sectors
Units: sectors of 1 * 4096 = 4096 bytes
Sector size (logical/physical): 4096 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
```

<4> mkfs -

```
gopika_arumugm@cloudshell:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.47.0 (5-Feb-2023)
/dev/sdb1 is apparently in use by the system; will not make a filesystem here!
gopika_arumugm@cloudshell:~$
```

<5> mount -

```
gopika_arumugm@cloudshell:~$ sudo mount /dev/sdb1 /mnt
gopika_arumugm@cloudshell:~$
```

<6> umount -

```
gopika_arumugm@cloudshell:~$ sudo umount /mnt
gopika_arumugm@cloudshell:~$
```

<7> blkid -

```
gopika_arumugm@cloudshell:~$ blkid
gopika_arumugm@cloudshell:~$ lsblk
```

<8> 1sblk -

```
gopika_arumugm@cloudshell:~$ lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda       8:0    0  100G  0 disk
├─sda1    8:1    0 95.8G  0 part /run/google/devshell
│                                /var/lib/docker
│                                /etc/ssh/keys
│                                /var/config/tmux
│                                /etc/resolv.conf
│                                /etc/hostname
│                                /etc/hosts
│                                /root
├─sda2    8:2    0   16M  0 part
├─sda3    8:3    0    2G  0 part
├─sda4    8:4    0   16M  0 part
├─sda5    8:5    0    2G  0 part
├─sda6    8:6    0  512B  0 part
├─sda7    8:7    0  512B  0 part
├─sda8    8:8    0   16M  0 part
├─sda9    8:9    0  512B  0 part
├─sda10   8:10   0  512B  0 part
├─sda11   8:11   0    8M  0 part
└─sda12   8:12   0   32M  0 part
sdb       8:16   0    5G  0 disk
└─sdb1    8:17   0    5G  0 part /home
gopika_arumugm@cloudshell:~$
```

<9> parted -

```
gopika_arumugm@cloudshell:~$ sudo parted -l
Model: Google PersistentDisk (scsi)
Disk /dev/sda: 107GB
Sector size (logical/physical): 512B/4096B
Partition Table: gpt
Disk Flags: pmbr_boot

Number  Start   End     Size    File system  Name        Flags
11      32.8kB  8421kB  8389kB               RWFW        bios_grub
 6      8421kB  8422kB  512B                 KERN-C      chromeos_kernel
 7      8422kB  8422kB  512B                 ROOT-C
 9      8422kB  8423kB  512B                 reserved
10      8423kB  8423kB  512B                 reserved
 2      10.5MB  27.3MB  16.8MB               KERN-A      chromeos_kernel
 4      27.3MB  44.0MB  16.8MB               KERN-B      chromeos_kernel
 8      44.0MB  60.8MB  16.8MB  ext4         OEM
12      128MB   161MB   33.6MB  fat16        EFI-SYSTEM  boot, legacy_boot, esp
 5      161MB   2309MB  2147MB               ROOT-B
 3      2309MB  4456MB  2147MB  ext2         ROOT-A
 1      4456MB  107GB   103GB   ext4         STATE


Model: Google PersistentDisk (scsi)
Disk /dev/sdb: 5369MB
Sector size (logical/physical): 512B/4096B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
 1      1049kB  5369MB  5368MB  primary  ext4
```

<10> tune2fs -

```
gopika_arumugm@cloudshell:~$ sudo tune2fs -l /dev/sda1
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name:   STATE
Last mounted on:          /var
Filesystem UUID:          9f2dbcb8-9534-4465-bea2-481409599ab8
Filesystem magic number:  0xEF53
Filesystem revision #:    1 (dynamic)
Filesystem features:      has_journal ext_attr resize_inode dir_index filetype needs_recovery ext
ent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags:         signed_directory_hash
Default mount options:    user_xattr acl
Filesystem state:         clean
Errors behavior:          Remount read-only
Filesystem OS type:       Linux
Inode count:              6258720
Block count:              25126395
Reserved block count:     0
Overhead clusters:        418418
Free blocks:              5634073
Free inodes:              4791567
First block:              0
Block size:               4096
Fragment size:            4096
Group descriptor size:    64
Reserved GDT blocks:      609
Blocks per group:         32768
Fragments per group:      32768
Inodes per group:         8160
Inode blocks per group:   510
Flex block group size:    16
Filesystem created:       Fri May  8 10:15:28 2026
```

CHAPTER – 8 SYSTEM MONITORING AND PERFORMANCE

<1> top -

```
top - 02:46:23 up  1:33,  0 user,  load average: 0.11, 0.13, 0.30
Tasks:  32 total,   1 running,  31 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.1 st
MiB Mem :  15998.9 total,  12060.0 free,   2245.9 used,   1972.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.  13753.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    779 gopika_+  20   0   11.3g 117648  51016 S   0.3   0.7   0:03.27 node
   1502 gopika_+  20   0   42.5g 790980  62128 S   0.3   4.8   0:33.76 node
   1784 gopika_+  10 -10   11916   5696   3496 R   0.3   0.0   0:00.03 top
      1 root      20   0    4324   3324   3040 S   0.0   0.0   0:00.07 bash
     10 syslog    20   0  222140   3872   2520 S   0.0   0.0   0:00.02 rsyslogd
     26 root      20   0   38768  28128  11024 S   0.0   0.2   0:00.65 python
     27 root      20   0    6196   2620   2380 S   0.0   0.0   0:00.00 logger
     68 root      10 -10   12024   4088   2972 S   0.0   0.0   0:00.01 sshd
    358 root      20   0 2096160  85708  61808 S   0.0   0.5   0:00.52 dockerd
    437 root      20   0 1234780  12528   8208 S   0.0   0.1   0:00.50 editor-proxy
    438 root      20   0   16328   6428   5500 S   0.0   0.0   0:00.01 sudo
    454 root      20   0 1226356   2796   1976 S   0.0   0.0   0:00.00 tmux-agent
    488 root      20   0 1867468  46380  32284 S   0.0   0.3   0:01.77 containerd
    687 root      20   0    2696   1528   1432 S   0.0   0.0   0:00.00 sleep
```

<2> htop -

```
    0[||                                          0.7%] Tasks: 32, 97 thr, 0 kthr; 1 running
    1[|                                           0.7%] Load average: 0.05 0.11 0.28
    2[||                                          0.7%] Uptime: 01:33:58
    3[||                                          1.0%]
  Mem[||||||||||||||||                     1.92G/15.6G]
  Swp[                                           0K/0K]

  Main   I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
   1789 gopika_aru  10 -10  8420  4812  3728 R   0.7  0.0  0:00.09 htop
      1 root        20   0  4324  3324  3040 S   0.0  0.0  0:00.05 /bin/bash /google/scripts/onrun.sh sleep infinity
     10 syslog      20   0  216M  3872  2520 S   0.0  0.0  0:00.00 /usr/sbin/rsyslogd
     11 syslog      20   0  216M  3872  2520 S   0.0  0.0  0:00.01 /usr/sbin/rsyslogd
     12 syslog      20   0  216M  3872  2520 S   0.0  0.0  0:00.00 /usr/sbin/rsyslogd
     13 syslog      20   0  216M  3872  2520 S   0.0  0.0  0:00.01 /usr/sbin/rsyslogd
     26 root        20   0 38768 28128 11024 S   0.0  0.2  0:00.67 /usr/bin/python /usr/bin/supervisord -n -c /google/
     27 root        20   0  6196  2620  2380 S   0.0  0.0  0:00.00 logger -t supervisord
     68 root        10 -10 12024  4088  2972 S   0.0  0.0  0:00.01 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/e
    358 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.20 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    408 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.05 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    409 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.06 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    410 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.07 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    411 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    412 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    435 root        20   0 2047M 85708 61808 S   0.0  0.5  0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460
    437 root        20   0 1205M 12528  8208 S   0.0  0.1  0:00.12 /google/devshell/editor/proxy/editor-proxy -static-
    438 root        20   0 16328  6428  5500 S   0.0  0.0  0:00.01 sudo /google/devshell/tmux-agent
```

<3> iostat -

```
gopika_arumugm@cloudshell:~$ iostat
Linux 6.6.137+ (cs-553674751103-default)        05/15/2026      _x86_64_        (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.74    0.00    0.40    0.10    0.01   98.75

Device             tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0              1.35        73.78         0.00         0.00     435468          0          0
sda               6.53       287.75        43.77         0.00    1698395     258349          0
sdb               0.17        22.68         0.26         0.00     133877       1564          0
```

<4> vmstat -

```
gopika_arumugm@cloudshell:~$ vmstat
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
 1  0      0 12351880  77832 1943164    0    0   320    42  821    3  1  0 99  0  0  0
gopika_arumugm@cloudshell:~$
```

<5> sar –

```
gopika_arumugm@cloudshell:~$ sar 1 5
Linux 6.6.137+ (cs-553674751103-default)        05/15/2026      _x86_64_        (4 CPU)

02:53:12 AM     CPU     %user     %nice   %system   %iowait    %steal     %idle
02:53:13 AM     all      0.25      0.00      0.25      0.00      0.00     99.50
02:53:14 AM     all      0.00      0.00      0.50      0.00      0.00     99.50
02:53:15 AM     all      0.50      0.00      0.25      0.00      0.00     99.25
02:53:16 AM     all      0.00      0.00      0.25      0.00      0.25     99.50
```

<6> free -

```
gopika_arumugm@cloudshell:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:            15Gi       2.2Gi        11Gi       1.1Mi       1.9Gi        13Gi
Swap:             0B          0B          0B
```

<7> mpstat -

```
gopika_arumugm@cloudshell:~$ mpstat
Linux 6.6.137+ (cs-553674751103-default)        05/15/2026      _x86_64_        (4 CPU)

02:50:52 AM  CPU    %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
02:50:52 AM  all    0.74    0.00    0.38    0.10    0.00    0.02    0.01    0.00    0.00   98.75
```

<8> uptime -

```
gopika_arumugm@cloudshell:~$ uptime
 02:49:39 up  1:36,  0 user,  load average: 0.05, 0.10, 0.25
```

CHAPTER – 9 USER MANAGEMENT

<1> useradd -

```
gopika_arumugm@cloudshell:~$ sudo useradd gopi
```

<2> userdel -

```
gopika_arumugm@cloudshell:~$ sudo userdel gopi
```

<3> usermod

```
gopika_arumugm@cloudshell:~$ sudo usermod -l gops gopi
gopika_arumugm@cloudshell:~$ 
```

<4> passwd

```
gopika_arumugm@cloudshell:~$ passwd
Changing password for gopika_arumugm.
Current password: 
```

<5> chage -

```
gopika_arumugm@cloudshell:~$ sudo chage -l gopika_arumugm
Last password change                                    : May 15, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
```

<6> groups -

```
gopika_arumugm@cloudshell:~$ groups
gopika_arumugm adm sudo docker
```

<7> groupadd -

```
gopika_arumugm@cloudshell:~$ sudo groupadd stud
gopika_arumugm@cloudshell:~$ groups
gopika_arumugm adm sudo docker
gopika_arumugm@cloudshell:~$ ^C
gopika_arumugm@cloudshell:~$ sudo usermod -aG stud gopika_arumugm
gopika_arumugm@cloudshell:~$ groups
gopika_arumugm adm sudo docker
gopika_arumugm@cloudshell:~$ newgrp stud
gopika_arumugm@cloudshell:~$ groups
stud adm sudo docker gopika_arumugm
gopika_arumugm@cloudshell:~$ 
```

<8> groupdel -

```
gopika_arumugm@cloudshell:~$ sudo groupdel stud
gopika_arumugm@cloudshell:~$ groups
groups: cannot find name for group ID 1003
```

<9> newgrp -

```
gopika_arumugm@cloudshell:~$ newgrp gopika_arumugm
gopika_arumugm@cloudshell:~$ 
```

<10> su -

```
gopika_arumugm@cloudshell:~$ su gopika
Password:
bash: cd: /home/gopika_arumugm: Permission denied
gopika@cloudshell:/home/gopika_arumugm$ ^C
```

CHAPTER – 10 COMPRESSING AND ARCHIVING

<1> bzip2 –

```
gopika_arumugm@cloudshell:~$ sudo chattr -i file.txt
gopika_arumugm@cloudshell:~$ echo hello > myfile.txt
gopika_arumugm@cloudshell:~$ bzip2 myfile.txt
gopika_arumugm@cloudshell:~$
```

<2> bunzip2 -

```
gopika_arumugm@cloudshell:~$ bunzip2 file.txt.bz2
bunzip2: Output file file.txt already exists.
gopika_arumugm@cloudshell:~$
```

<3> xz -

```
gopika_arumugm@cloudshell:~$ xz file.txt
gopika_arumugm@cloudshell:~$
```

<4> unxz -

```
gopika_arumugm@cloudshell:~$ unxz file.txt.xz
gopika_arumugm@cloudshell:~$
```

<5> tar -

```
gopika_arumugm@cloudshell:~$ tar -cvf archive.tar file.txt
file.txt
gopika_arumugm@cloudshell:~$
```

<6> zip -

```
gopika_arumugm@cloudshell:~$ zip archive.zip file.txt
  adding: file.txt (stored 0%)
```

<7> unzip -

```
gopika_arumugm@cloudshell:~$ unzip archive.zip
Archive:  archive.zip
replace file.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
 extracting: file.txt
gopika_arumugm@cloudshell:~$
```

<8> 1zma -

```
gopika_arumugm@cloudshell:~$ lzma file.txt
```

<9> 1zop –

```
gopika_arumugm@cloudshell:~$ lzop -d file.txt.lzo
lzop: file.txt: already exists; not overwritten
skipping file.txt.lzo [file.txt]
gopika_arumugm@cloudshell:~$
```

<10> unlzop -

```
gopika_arumugm@cloudshell:~$ lzop file.txt
gopika_arumugm@cloudshell:~$ unlzop file.txt.lzo
```